Nikhil Vanjani

Research Interests

Cryptography, Blockchains

Education

Carnegie Mellon University (CMU) — Pittsburgh, PA, USA
Ph.D., Electrical and Computer Engineering — Jan 2022 - Dec 2025
- Advisor: Elaine Shi
- Thesis: *Functional Encryption and its Applications to Decentralized Systems*

M.S., Information Security — Aug 2020 - Dec 2021
- Advisor: Elaine Shi
- Thesis: *Multi-Input Inner Product Encryption: Function-Hiding Instantiations without Random Oracles*

Indian Institute of Technology Kanpur (IITK) — Kanpur, UP, India
B.Tech., Computer Science and Engineering — Jul 2014 - May 2018

Professional Experience

Babylon Labs | Senior Researcher — Jan 2026 - Present
Supervisor: David Tse
Leading the design and security analysis of practically efficient SNARK Proof Verification Systems on Bitcoin

NTT Research | Research Intern — Jun - Aug 2025
Supervisor: Pratish Datta
Led the design and implementation of practically efficient and post-quantum secure attribute-based encryption schemes, advancing the practicality of lattice-based cryptography

0xPARC Foundation | Research Engineer — Mar - May 2025
Supervisor: Brian Lawrence
Benchmarked modern zero-knowledge proof systems (Plonky2/Plonky3), providing performance insights to guide practical adoption of advanced cryptographic protocols

NTT Research | Research Intern — Jun - Aug 2024
Supervisor: Pratish Datta
Developed foundational advances in multi-authority attribute-based encryption by proving full adaptive security for the classic Lewko–Waters scheme and designing the first scheme for Arithmetic Span Programs

Algorand | Smart Contracts Research Intern — May - Aug 2021
Supervisor: Jing Chen — Code: BLS12-381 ops, BLS signatures
Implemented EIP-2537 compliant BLS12-381 curve ops and BLS signatures, and integrated them into Algorand's smart-contract stack, significantly accelerating aggregate signature verification in consensus.

IIT Madras | Research Assistant — Aug 2019 - Jun 2020
Supervisor: Shweta Agrawal
Designed a blockchain-based voting system with support for vote verification

Cohesity | Member of Technical Staff — Jun 2018 - Jul 2019
Built and integrated authentication, data deduplication, multistreaming features in distributed backup systems

Publications

Unless otherwise noted, the author order is either alphabetical or randomized.

Conference Proceedings

[6] **Large-Universe (Multi-Authority) ABE from LWE**
Pratish Datta, Yannis Rouselakis, Junichi Tomida, Nikhil Vanjani
ACM CCS 2026

[5] **New Constructions of Functional Adaptor Signatures: Broader Functions and Improved Efficiency**
Nikhil Vanjani (first author), Garrett Greiner (first author), Sri AravindaKrishnan Thyagarajan, Pratik Soni
IEEE Security and Privacy (Oakland) 2026 Code

[4] **Fully Adaptive Decentralized MA-ABE: Simplified, Optimized, ASP Supported**
Pratish Datta, Junichi Tomida, Nikhil Vanjani
IACR Asiacrypt 2025 Paper

[3] **Functional Adaptor Signatures: Beyond All-or-Nothing Blockchain-based Payments**
Nikhil Vanjani (first author), Pratik Soni, Sri AravindaKrishnan Thyagarajan
ACM CCS 2024, TPMPC 2025 Code, Paper

[2] **Non-Interactive Anonymous Router with Quasi-Linear Router Computation**
Rex Fernando, Elaine Shi, Pratik Soni, Nikhil Vanjani, Brent Waters
IACR TCC 2023 Paper

[1] **Multi-Client Inner Product Encryption: Function-Hiding Instantiations Without Random Oracles**
Elaine Shi, Nikhil Vanjani
IACR PKC 2023 Paper

Maunscripts

[2] **Non-Interactive Translation of Winternitz Signatures to Lamport Signatures via Secret Sharing**
Mikhail Sergeevitch, Konrad Staniec, David Tse, Nikhil Vanjani, Robin Linus Woll
In Submission Paper

[1] **Unbounded Large-Universe Decentralized MA-ABE from Static Assumptions**
Pratish Datta, Junichi Tomida, Nikhil Vanjani
In Submission

Patents

[1] **Multi-Authority Attribute-Based Encryption with Adaptive Security for Arithmetic Span Programs**
Pratish Datta, Junichi Tomida, Nikhil Vanjani
US Patent App. **63/875,152**, Filed Sep 3, 2025 (Pending)

Scholastic Achievements

- **Academic Research Grant**, Stellar Development Foundation 2024
  Principal Investigators: Pratik Soni, Sri AravindaKrishnan Thyagarajan
- **Carnegie Institute of Technology Dean's Fellowship** 2022
- **Best Masters Thesis**, Information Networking Institute, CMU 2022
- **Tuition Scholarship ($9000)**, Information Networking Institute, CMU 2020
- **Red Hat Certified System Administrator (RHCSA)**, Certificate Number: 170-124-598 2017
- **Rank 1**, Blockchain Hackathon at Techkriti, IIT Kanpur 2017
- **Rank 461**, Codechef Snackdown (Final Round) 2015
- **All India Rank 201**, Joint Entrance Examination (JEE) Advanced 2014

Professional Service

- **Program Committee:**
  - Crypto Valley Conference 2025
- **External Reviewer:**
  - Crypto (2024, 2025, 2026); Eurocrypt (2024); Asiacrypt (2022); TCC (2023, 2024); Indocrypt (2024)
  - CCS (2024, 2026); FC (2024, 2025); TDSC (2023)
  - STOC (2026)
- **Co-organizer of CMU Cylab Crypto Seminar** 2023 - 2025

Teaching / Mentoring

- **Foundations of Blockchains (15435), CMU** | *Teaching Assistant* — Sep - Dec 2022, Sep - Dec 2023
- **Intro to Information Security (14741), CMU** | *Teaching Assistant* — Feb - May 2021
- **Blockchains, Association of Computing Activities, IITK** | *Student Instructor* — Jan - Apr 2018
- **Cryptography, Association of Computing Activities, IITK** | *Student Instructor* — Aug - Nov 2017
- **Cyber Security, Association of Computing Activities, IITK** | *Student Instructor* — Jan - Apr 2017

Selected Talks

- **New Constructions of Functional Adaptor Signatures: Broader Functions and Improved Efficiency.**
  IEEE Security and Privacy (Oakland) conference — Slides | May 2026
- **Functional Fair Exchange: Beyond All-or-Nothing Data Exchange.**
  Boston University Security Seminar — Slides | Feb 2026
  Yale Applied Cryptography Lab Seminar — Slides | Jan 2026
- **Fully Adaptive Decentralized MA-ABE: Simplified, Optimized, ASP Supported.**
  UC Berkeley Crypto Seminar — Slides | Nov 2025
  Stanford Security Seminar — Slides | Oct 2025
  NTT Research CIS Seminar — Slides | Oct 2025
  CMU Crypto Seminar — Slides | Sep 2025
- **Functional Adaptor Signatures: Beyond All-or-Nothing Blockchain-based Payments.**
  Invited Lecture, University of Utah — Slides | Oct 2024
  ACM CCS conference — Slides | Oct 2024
- **Non-Interactive Anonymous Router with Quasi-Linear Router Computation**
  IACR TCC conference — Slides | Dec 2023
  Ph.D. Qualifying Exam, CMU — Slides | Nov 2022
- **Multi-Client Inner Product Encryption: Function-Hiding Instantiations Without Random Oracles**
  IACR PKC conference — Slides | May 2023
  CMU Theory Lunch — Slides | Apr 2023
  MS thesis defense, CMU — Slides | Nov 2021
- **Attribute-based Signatures for Unbounded Circuits in the Random Oracle Model**
  Cryptography reading group talk, IITM — Slides | Jul 2020
- **Obfuscation of Probabilistic Circuits and Applications**
  Course project for Computing on Encrypted Data, IITM — Slides | Nov 2019
- **Two case studies on advances in Blockchains: Algorand, Zcash**
  Seminar talk for National Blockchain Project being undertaken by C3I Center, IITK — Slides | Apr 2018

Personal Information

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- Google Scholar: https://scholar.google.com/citations?user=TgFRe-YAAAAJ
- Github: https://github.com/nikhilvanjani
- LinkedIn: https://www.linkedin.com/in/nikhilvanjani/
- Website: https://nikhilvanjani.github.io